Hongming Qiu (Henry)

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Education

University of Illinois Urbana-Champaign

M.S. in Electrical and Computer Engineering (Thesis) — *Expected May 2028*

B.S. in Computer Engineering | Overall GPA: 3.99/4.00 | Highest Honors | Bronze Tablet — *August 2026*

Research Experience

Kernel Security Researcher — **September 2025 – Present**

UIUC | Advisor: Prof. Tianyin Xu | In collaboration with IBM Research [code]

- Designed and implemented Linux IMA extensions for policy-controlled **eBPF measurement and appraisal**, including TPM-backed attestation, PKCS#7 signature verification, and policy predicates over program and attachment types.
- Built **race-safe reappraisal and revocation** of resident eBPF programs using retained integrity metadata and RCU; removed references across links, pins, file descriptors, tail-call maps, and cgroup attachments.
- Integrated signed loading and recovery with **Cilium**; evaluated vulnerable datapath remediation and a concealed eBPF rootkit with zero residual programs across 34 purge configurations; wrote the resulting manuscript as first author.
- Contributed to ongoing research evaluating flexible **Linux kernel control-flow integrity** enforcement with eBPF by reproducing OS-CFI's origin-sensitive design.

LLVM-Based Static Malware Detection Research — **February 2025 – Present**

Independent Research | Three-person team [code]

- Co-developed an LLVM-based polymorphic compilation framework for evaluating **static antivirus evasion** across Windows and Linux targets.
- Designed the original **EliminateIAT** pass, rewriting imported calls through runtime LoadLibrary/GetProcAddress stubs while preserving LLVM invoke control flow and variadic call sites; led MSVC/LLVM toolchain integration for complex Windows codebases.
- Designed **Masquerade** to copy sections and imports from benign Windows programs and keep them from being removed during linking; improved string obfuscation to support complex access patterns.
- Evaluated 1,160 artifacts from seven malware families against up to 76 VirusTotal engines; measured **95.5% mean inter-build byte divergence** and sub-1% mean detection for the strongest pass combinations.

Selected Systems Projects

Out-of-Order RISC-V Processor | *SystemVerilog, ECE 411, Team of 3* — **Spring 2026**

- Co-designed and verified a **2-way superscalar RV32IM processor** with explicit register renaming, split load/store queues, store forwarding, pipelined caches, and early branch recovery.
- Implemented **gshare branch prediction and return-address prediction**, reducing branch mispredictions from about **60% to 10%**; sustained **over 1 IPC** on compression and ray-tracing benchmarks at 500 MHz within a 300,000 μm^2 area limit.

RISC-V Operating System Kernel | *C, RISC-V Assembly, ECE 391* — **Fall 2024**

- Built a RISC-V kernel with process management, pipe-based inter-process communication, and custom device drivers.

Teaching & Leadership

ECE 391 Course Assistant — **January 2025 – Present**

ECE Department, University of Illinois Urbana-Champaign

- Mentored **20+ students weekly** in a class of 150+, debugging RISC-V operating systems spanning process management, pipe-based inter-process communication, and custom device drivers.
- Adapted Doom and Rogue to run on the course's RISC-V operating system as end-to-end workloads for student kernels.

Admin & Purple Team Lead — **September 2023 – Present**

SIGPwny, cybersecurity club at the University of Illinois Urbana-Champaign

- Authored **25+ reverse-engineering and exploitation challenges** for UIUCTF and SIGPwny; led advanced security workshops for 50+ club members.
- Built and administered a Proxmox security range with **20+ VMs**, segmented attacker and practice networks, a pfSense firewall, and VPN access.

Technical Skills

Languages: C, C++, Python, Bash, x86 and RISC-V Assembly, SystemVerilog
Systems: Linux kernel, eBPF, LLVM, IMA, TPM, RCU, Cilium, operating systems, container networking
Security: Program analysis, reverse engineering, binary exploitation, exploit development
Certifications: OSED, OSCP+, Certified CyberDefender (CCD)